How Predictable Are Rare Precipitation Extremes? A Calibration-Aware Assessment for London and Paris

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Abstract

How predictable are rare precipitation extremes, and how should that predictability be measured? We address both questions using thirty years (1989–2018) of a standardised precipitation-extreme index for London (maritime) and Paris (more continental), with strictly antecedent ERA5 drivers. We argue rare-event forecasts must be judged by *calibration-aware proper scores* against climatology — not discrimination metrics or conformal coverage, which alone can mislead — and assemble a protocol (Brier decomposition, threshold-weighted CRPS skill, bootstrap intervals, multi-split checks) guarding against the single-seed and quantile-selection pitfalls that inflate apparent skill. Across a bake-off from logistic regression to a deep Gaussian-process extreme-value hurdle, we find — consistently in both cities — that (i) **simple calibrated models outperform the elaborate ones**, and the response distribution matters more than model class; (ii) **occurrence** is weakly but genuinely predictable from antecedent convective instability and circulation; and (iii) **the magnitude of the largest events** is weakly but robustly predictable from convective instability (CAPE/CIN), with a CRPS skill over climatology excluding zero across four independent splits per city. The effects are small, statistically robust, and physically plausible; replication across two cities that share the same reanalysis source offers modest, not decisive, evidence against a location-specific artefact.

Keywords: extreme-value theory; proper scoring rules; forecast verification; conformal prediction; Gaussian processes; convective instability.

1 Introduction

Extreme precipitation is among the most damaging natural hazards facing temperate cities. Short, intense rainfall episodes trigger surface-water and river flooding, disrupt transport, overwhelm drainage, and impose large and unevenly distributed economic and social costs. In cities such as London and Paris such events occur on only a small fraction of days, yet a single episode can cause disproportionate harm, and both observational records and climate projections indicate that the frequency and intensity of heavy rainfall are increasing (IPCC, 2021). Skilful and, above all, *trustworthy* early warning of these events would let emergency managers, infrastructure operators, and insurers act pre-emptively. This motivates probabilistic forecasts of extreme precipitation at horizons of days to weeks ahead.

The statistical obstacles are severe. Extreme days are rare — typically 1–5% of the record — so a model that simply predicts “no extreme” is accurate yet useless, and conventional accuracy is the wrong yardstick. The daily precipitation index is zero-inflated, strongly right-skewed, and heavy-tailed, violating the Gaussian assumptions underlying standard regression. The observational record is short relative to the rarity of the events, leaving few positive examples from which to learn, and the day-to-day persistence of extremes is limited, so intrinsic predictability is modest. Together these features cap raw forecast *skill*, a ceiling any honest treatment must confront rather than obscure.

We argue that, precisely because raw predictability is limited, the decisive quality of a forecast in this setting is not discrimination but *calibration*: whether the issued probabilities can be trusted. A forecast that assigns a 5% probability to an extreme is useful for decision-making only if extremes do occur on about 5% of such days; an overconfident forecast that systematically overstates rare-event probabilities causes false alarms and wasted resources, while an underconfident one breeds complacency. The appropriate goal is therefore to *maximise sharpness subject to calibration* (Gneiting et al., 2007): to be as informative as the data allow while remaining honest. Calibration matters most exactly where it is hardest to achieve — in the tail — and evaluating it there is itself subtle, as the *forecaster's dilemma* makes clear (Lerch et al., 2017).

No existing methodology delivers calibrated, guaranteed tail probabilities for rare-event forecasting, because the three relevant tools each supply only part of the answer. *Extreme-value theory* (EVT) furnishes asymptotically justified tail models, but is most often used descriptively — for return levels or non-stationary tail regression — rather than for flexible, covariate-driven probabilistic forecasting, and it offers no finite-sample coverage guarantee (Coles, 2001; Davison and Smith, 1990). *Flexible machine learning*, including deep models and Gaussian processes, captures non-linear dependence on atmospheric predictors and yields predictive distributions, but those distributions are typically miscalibrated in the tail and come with no guarantee (Galib et al., 2022; Lakshminarayanan et al., 2017). *Conformal prediction* provides distribution-free, finite-sample coverage, but its standard form under-covers the tail — the relevant high quantile cannot be resolved from a finite calibration set — and assumes exchangeability, which daily time series violate (Gibbs and Candès, 2021; Pasche et al., 2025). Moreover, essentially all two-part (hurdle) forecasters assume that the *occurrence* of an extreme and its *intensity* are conditionally independent, discarding the abundant occurrence data that could inform the data-starved tail.

Rather than advocate a single model, we therefore ask a more basic pair of questions — *how predictable are these extremes, and how should that predictability be measured?* — and answer them with a careful evaluation protocol and a bake-off of models spanning logistic regression, zero-inflated distributional regressions (log-normal, gamma, and the threshold-free extended generalised Pareto distribution), and a deep Gaussian-process extreme-value hurdle with conformal calibration.

Contributions.

1. A **calibration-aware evaluation protocol** for rare-event precipitation forecasting: Brier reliability/resolution decomposition, threshold- and quantile-weighted CRPS, skill scores against climatology, bootstrap confidence intervals, and multi-split robustness — designed to resist the single-seed and quantile-selection artefacts that inflate apparent skill, and to demonstrate that conformal coverage is insufficient as a stand-alone measure of forecast quality.

2. A **rigorous predictability assessment, replicated in two contrasting cities**: occurrence of an extreme is weakly but genuinely predictable from antecedent state; and the *magnitude* of the largest events carries weak but robust predictability tied to antecedent **convective instability** (CAPE/CIN), with a CRPS skill over climatology whose moving-block bootstrap interval excludes zero across four independent train/test splits in each city — and is stronger in the more convective Paris, consistent with a meteorological rather than a purely local origin (though two cities in one climate zone offer only modest evidence on geographic generality).

3. A **cautionary methodological finding**: across the bake-off, *simpler calibrated models (logistic regression, gradient-boosted trees) outperform the elaborate deep-GP–EVT model*,

the choice of response distribution dominates model class, and a sensitivity analysis shows it is the added *depth* that fails on this low-signal problem — a plain Gaussian process is competitive.

Empirically, we study the daily standardised precipitation index over 1989–2018 for two cities of contrasting climate: London (maritime, 417 extreme days at the 95th percentile) and Paris (more continental and convective, 607 extreme days). The 95th-percentile index is the primary target; the rarer 99th-percentile index (137 events for London) is too data-scarce for the held-out intensity analysis (its occurrence-only results are in the Supplementary Material). Predictors are strictly antecedent ERA5 atmospheric drivers (circulation, moisture, and convective-instability fields) and large-scale teleconnection indices. Because the index is itself an ERA5-derived precipitation product, we exclude all precipitation fields from the predictors and lag every driver, so the assessment is of genuine antecedent predictability rather than diagnosis.

The remainder of the paper is organised as follows. Section 2 positions our work within the literatures on extreme-value modelling, Gaussian processes, conformal prediction, and forecast evaluation. Section 3 describes the data and predictors. Section 4 sets out the models we compare, the conformal calibration, and the evaluation protocol. Section 5 presents the results, and Sections 6–7 discuss implications and conclude.

2 Related work

Our work sits at the intersection of four literatures; we review each in turn and then state precisely how this study differs.

Statistical modelling of precipitation extremes. Extreme-value theory provides the asymptotic foundation for tail modelling. The peaks-over-threshold approach models exceedances above a high threshold by the Generalised Pareto distribution (Pickands, 1975; Davison and Smith, 1990), the framework we adopt for the intensity component; the block-maxima approach uses the Generalised Extreme Value distribution (Coles, 2001). EVT is widely applied in hydrology and climatology (Katz et al., 2002), including non-stationary and spatial Bayesian formulations of precipitation extremes (Cooley et al., 2007). This body of work is, however, predominantly *descriptive* — estimating return levels or letting tail parameters vary smoothly with a few covariates — rather than producing flexible, covariate-rich *forecasts* at a fixed lead time, and it offers no finite-sample coverage guarantee. We retain the GPD tail but embed it in a flexible predictive model and calibrate it conformally.

Machine learning and deep learning for extremes. Data-driven methods increasingly target extremes: Galib et al. (2022) forecast block maxima with a deep network constrained by GEV theory, and neural networks have been used to estimate extreme-value parameters. More broadly, predictive uncertainty in deep models is quantified with deep ensembles (Lakshminarayanan et al., 2017) and Monte-Carlo dropout (Gal and Ghahramani, 2016). These methods capture non-linear dependence on predictors and return uncertainty estimates, but they are typically miscalibrated in the tail and lack distribution-free guarantees — the gap our conformal layer closes.

Gaussian processes: depth and multiple outputs. Gaussian processes offer principled non-parametric regression with built-in uncertainty (Rasmussen and Williams, 2006), made scalable through sparse variational inference with inducing points (Titsias, 2009; Hensman et al., 2013). Expressiveness is increased by composing layers in deep Gaussian processes (Damianou and Lawrence, 2013), trained by doubly-stochastic variational inference (Salimbeni and Deisenroth, 2017), or by learning the kernel through a neural feature map in deep kernel learning (Wilson et al., 2016). Correlated outputs are modelled by multi-output GPs and coregionalisation (Bonilla et al., 2008; Álvarez et al., 2012). A GP with a Gaussian likelihood cannot represent heavy-tailed, skewed, zero-inflated responses; we therefore pair the GP prior with an EVT likelihood and use coregionalisation not to share information across tasks but to couple the two halves of a hurdle model.

Two-part, hurdle, and selection models. Semicontinuous and zero-inflated outcomes are commonly modelled by two-part (hurdle) decompositions separating occurrence from intensity. Allowing the two parts to be *dependent* has a long history: correlated random-effects formulations (Olsen and Schafer, 2001), copula couplings, and the sample-selection models of econometrics (Heckman, 1979), which explicitly address the correlation between a selection (occurrence) equation and an outcome (intensity) equation. To our knowledge this dependence has not been exploited within a deep-GP–EVT *forecasting* model, where — because intensity is observed only on the rare extreme days — borrowing strength from the abundant occurrence data is especially valuable.

Conformal prediction. Conformal prediction yields distribution-free, finite-sample coverage for any underlying predictor (Vovk et al., 2005). Split-conformal methods make this practical for regression (Lei et al., 2018), and conformalised quantile regression adapts the intervals to heteroscedasticity (Romano et al., 2019). Exchangeability can be relaxed for covariate shift (Tibshirani et al., 2019) and, crucially for us, for time series and online distribution shift through adaptive conformal inference (Gibbs and Candès, 2021; Xu and Xie, 2021). Coverage at very high confidence levels — the extreme regime — is addressed by Pasche et al. (2025), who combine EVT with conformal prediction to build reliable intervals for high-impact events. That work, the closest to ours, constructs *regression intervals* from a black-box extreme-quantile estimator under (weighted) exchangeability; it uses neither a Gaussian-process nor a generative model, does not forecast exceedance *probabilities*, and does not model temporal dependence natively. Here, by contrast, tail-targeted and temporally-adaptive conformal calibration is applied to a generative model, within a broader predictability study.

Forecast evaluation of extremes. Probabilistic forecasts are assessed with proper scoring rules (Gneiting and Raftery, 2007) under the paradigm of maximising sharpness subject to calibration (Gneiting et al., 2007); the Brier score admits a reliability–resolution decomposition (Murphy, 1973). For extremes, threshold- and quantile-weighted scoring rules focus evaluation on the tail while remaining proper (Gneiting and Ranjan, 2011), and naive restriction to extreme observations can paradoxically reward miscalibrated forecasts — the forecaster's dilemma (Lerch et al., 2017). Our evaluation is built on these principles.

Positioning. Much of this literature proposes ever more expressive models for extremes. We take the complementary stance of *measuring* how much is predictable and which modelling choices matter, under an evaluation protocol built from proper scoring rules, skill against climatology, and bootstrap and multi-split robustness. In doing so we find — against the prevailing direction — that under this dataset and evaluation framework simple calibrated models outperform an elaborate deep-GP extreme-value model, and that conformal coverage, increasingly reported as a calibration guarantee, is insufficient on its own as a measure of forecast quality. This emphasis on rigorous predictability assessment, rather than a new architecture, is the gap we fill.

3 Data

3.1 The precipitation-extreme index

Our response variable is a standardised daily precipitation-extreme index, computed for two European cities — **London** and **Paris** — derived from the Copernicus Climate Data Store product *Extreme precipitation risk indicators for Europe and European cities (1950–2019).* The underlying field is the ERA5 reanalysis dynamically downscaled to a 2 km × 2 km grid (ERA5-2km) with urban parameterisation, which resolves London as 53 and Paris as 157 grid cells. For each cell and day the indicator is the standardised exceedance

$$\mathcal{I} = \frac{P}{P_q}, \tag{1}$$

where P is daily precipitation and P_q is the fixed qth percentile of wet days ($P \geq 1\,\mathrm{mm}$) over the 1950–2019 reference period. The index is dimensionless, and $\mathcal{I} > 1$ marks an extreme: precipitation exceeding the historical qth-percentile level. To obtain a single city-level series we take the daily maximum of $\mathcal{I}$ over the city's cells, so an extreme day is one on which *any* part of the city exceeds its threshold. We analyse the thirty-year record 1989–2018 at the $q = 95$ (primary) and $q = 99$ thresholds. The two cities are chosen for their contrasting climates: London's maritime regime versus Paris's more continental, more convective one — a deliberate test of whether conclusions generalise.

3.2 Exploratory characteristics

The series comprises 10,957 daily observations. Its defining features — which directly motivate the modelling choices of Section 4 — are summarised in Table 1 and Figure 1, and are shared by both cities. First, extremes are *rare*: 3.8% of London days and 5.5% of Paris days exceed the 95th-percentile threshold (and only 1.3% at the 99th in London). Second, the index is *zero-inflated and heavy-tailed*: about a quarter of days are exactly zero, the median is below 0.02, yet the maximum exceeds 6 at the 95th-percentile standardisation — a strong right skew incompatible with a Gaussian response. Third, occurrence is *seasonal*: the monthly extreme rate peaks in summer (July–August, reaching 6% at the 95th percentile and 3% at the 99th) and is lowest in late winter and early spring, consistent with convective summer rainfall. Fourth, the extreme indicator exhibits only *limited temporal autocorrelation*, so the persistence available to an autoregressive forecaster is modest — one reason that drivers beyond the index's own history (Section 3.3) are essential.

3.3 Antecedent predictors

All predictors are *strictly antecedent*: for a horizon-h forecast of day $t + h$ they are functions only of information available up to day t. We use three groups.

1. **Index history.** Lagged values of the index (up to 14 days), rolling means and maxima over 7/14/30-day windows, counts of recent exceedances, and harmonic encodings of day-of-year and month to capture seasonality.
2. **ERA5 atmospheric drivers.** Antecedent fields from the ERA5 reanalysis over a synoptic-scale window around each city: mean sea-level pressure and its gradients, geopotential height (500/850 hPa), specific and total-column water vapour, integrated water-vapour transport, near-surface temperature and winds, and convective-instability diagnostics — convective available potential energy and inhibition, the K-index, and the Total-Totals index. Fields are aggregated to daily statistics (means, and maxima for the convective indices) and reduced to regional summaries.

Table 1: Summary characteristics of the 95th-percentile precipitation-extreme index (1989–2018; $N = 10{,}957$ days) for the two cities. Extreme days are those with $\mathcal{I} > 1$. Paris is more convective: more cells, more extremes, a heavier tail. (London's 99th-percentile statistics — 137 extreme days — are used only in the Supplementary Material.)

	London (maritime)	**Paris** (continental)
Grid cells	53	157
Extreme days (total)	417	607
Extreme rate (%)	3.81	5.54
train 1989–2008 (7,305 d)	287 (3.93%)	396 (5.42%)
test 2009–2018 (3,652 d)	130 (3.56%)	211 (5.78%)
Days equal to zero (%)	25.5	28.7
Median index	0.015	0.012
Maximum index	6.46	8.91
Peak season	July–August	

3. **Teleconnection indices.** Lagged North Atlantic Oscillation, Arctic Oscillation, and an El Niño–Southern Oscillation index, capturing large-scale circulation states relevant at longer horizons.

In total $d = 49$ predictors enter the full model: 20 index-history features (lags, rolling summaries, and seasonality harmonics), 26 ERA5 circulation, moisture, and convective-instability fields, and 3 teleconnection indices. In the analysis the drivers are introduced *cumulatively* — index history; + teleconnections; + ERA5 circulation and moisture; + convective instability — so that each increment isolates a class of predictors. All features are standardised using training-set statistics only; the complete list and the retrieval recipe are in Appendix A.

Remark 1 (No leakage; perfect-prognosis framing)**.** *ERA5 is a reanalysis that assimilates observations, so using fields at or after the target day would be circular. We therefore use only* lagged *drivers, known strictly before the forecast time. This is the standard* perfect-prognosis *predictability setting; an operational system would replace antecedent ERA5 with numerical-weather-prediction* forecast *fields, which we leave to future work (Section 7).*

3.4 Threshold, training, and testing

We take the 95**th-percentile index as the primary target**: its 417 extreme days (287 train, 130 test) admit the full analysis, including the held-out intensity test, which requires extreme days in both training and test across several splits. The rarer 99th-percentile index has only 137 extreme days in thirty years (101 train, 36 test) — too few for the intensity analysis — so its occurrence-only results are given in Supplementary Section S1 (Table S1). Unless stated, results are for the 95th percentile at horizon $h = 1$.

Because the data are a time series, all splits respect chronology. The primary split trains on 1989–2008 (7,305 days) and holds out 2009–2018 (3,652 days); preprocessing statistics and conformal calibration sets are derived from the training period only. To verify that results are not an artefact of a single test window, the intensity analysis is additionally repeated over four forward-only, expanding-window splits (test periods beginning 2003, 2006, 2009, and 2012), and all stochastic results are averaged over random seeds.

4 Methodology

Section 4.1 fixes notation and the predictability target. Section 4.2 describes the ladder of models we compare, from climatology and logistic regression through zero-inflated distributional

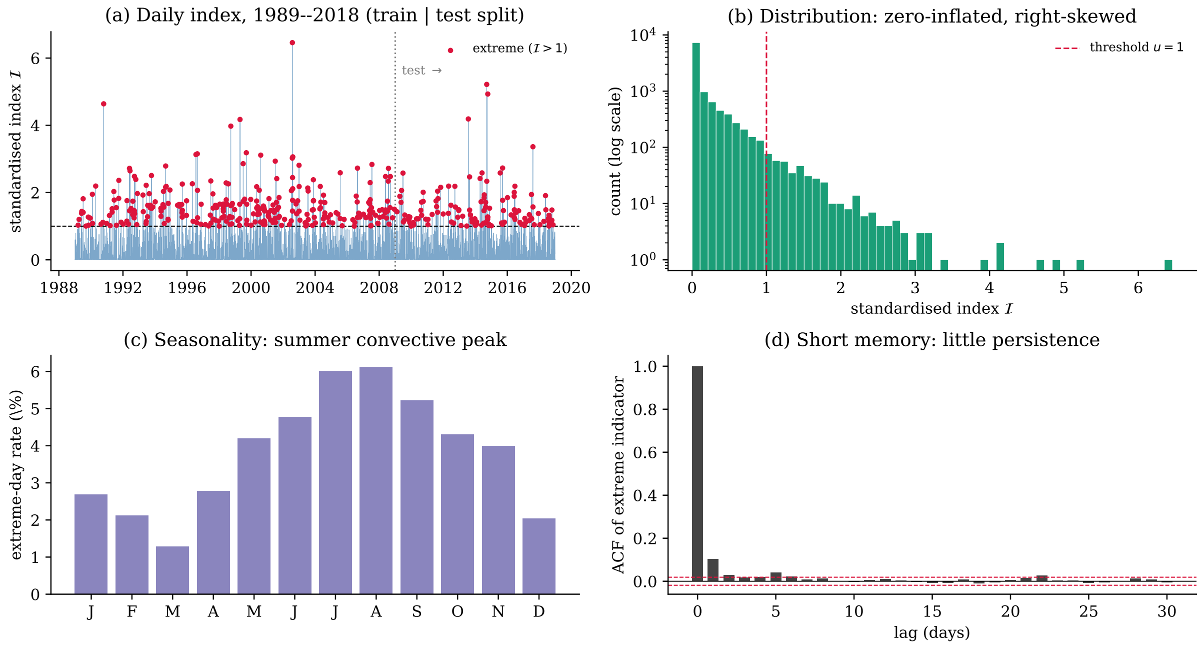

Figure 1: The London 95th-percentile precipitation-extreme index, 1989–2018 (Paris is qualitatively similar; see Supplementary Figure S1). **(a)** Daily series with extreme days ($\mathcal{I} > 1$) marked and the train/test split; **(b)** the distribution is zero-inflated ($\sim$a quarter of days are exactly zero) and strongly right-skewed (log count); **(c)** extremes peak in summer (convective rainfall); **(d)** the extreme indicator has very short memory — the autocorrelation is negligible beyond a day or two (dashed: 95% band) — which caps how much an autoregressive forecaster can achieve.

regressions to a deep Gaussian-process extreme-value hurdle. Section 4.3 describes the conformal calibration we assess (and why its coverage guarantee is, on its own, uninformative). Section 4.4 sets out the evaluation protocol — proper scores, skill against climatology, a hold-out intensity test, and bootstrap and multi-split robustness — which is itself a contribution.

4.1 Problem set-up and notation

Let $\mathcal{I}_t \geq 0$ denote the daily standardised precipitation index on day t, and let $u > 0$ be a fixed exceedance threshold (we take $u = 1$, the value at which the index marks an extreme, and report sensitivity to u in Section 5.1). Because $\mathcal{I} = P/P_q$ is itself standardised against the training-period qth percentile (Section 3.1), u and q are linked but distinct: varying q redefines the standardisation (a different, rarer event, as for the 99th-percentile analysis of Section 3.4), whereas varying u at fixed q asks how much of the standardisation ratio itself must be exceeded to count as extreme — e.g. $u = 1.25$ requires exceeding 125% of the qth-percentile level. A day is *extreme* when $\mathcal{I}_t > u$. Write the occurrence indicator and the excess as

$$O_t \;=\; \mathbf{1}\{\mathcal{I}_t > u\}, \qquad Y_t \;=\; (\mathcal{I}_t - u)\,\big|\,\{\mathcal{I}_t > u\}, \tag{2}$$

so that $O_t \in \{0, 1\}$ is observed every day while the excess Y_t is *defined and observed only on extreme days*. This asymmetry — many occurrence observations, few intensity observations — shapes both the modelling (Section 4.2) and the separate evaluation of occurrence and intensity predictability (Section 4.4).

Each day carries a vector of covariates $\boldsymbol{x}_t \in \mathbb{R}^d$ that are *strictly antecedent*: functions of information available up to and including day t (autoregressive lags and rolling summaries of the index, plus lagged ERA5 atmospheric drivers and teleconnection indices; see Section 3). Let $\mathcal{F}_t$ be the filtration generated by all information up to day t. For a forecast horizon $h \geq 1$ the target is the *predictive distribution* of $\mathcal{I}_{t+h}$ given $\mathcal{F}_t$, summarised by its conditional CDF

$$F_{t+h}(z) \;=\; \mathbb{P}(\mathcal{I}_{t+h} \le z \,|\, \mathcal{F}_t)\,, \qquad z \ge 0. \tag{3}$$

Two functionals are of primary interest: the *exceedance probability* $\pi_{t+h} = \mathbb{P}(\mathcal{I}_{t+h} > u \mid \mathcal{F}_t) = 1 - F_{t+h}(u)$ and the conditional law of the excess $Y_{t+h} \mid \{\mathcal{I}_{t+h} > u\}$.

4.2 Models compared

We assess a ladder of models of increasing complexity, all sharing the same strictly antecedent covariates $\boldsymbol{x}_t$ and judged by the protocol of Section 4.4. The aim is not to crown one architecture but to learn *how much* of the index is predictable and *which* modelling choices matter.

Baselines. *Climatology* issues the empirical distribution of the training index, the reference against which every skill score is measured; a *stationary peaks-over-threshold* (POT) model adds a Generalised Pareto tail fitted to the training excesses but no covariate dependence.

Occurrence: logistic regression and gradient-boosted trees. For the exceedance probability $\pi_{t+h} = \mathbb{P}(\mathcal{I}_{t+h} > u \mid \mathcal{F}_t)$ we fit a regularised logistic regression on $\boldsymbol{x}_t$ — the natural calibrated occurrence model — and, as a strong non-linear machine-learning baseline, an isotonic-calibrated gradient-boosted-tree classifier (XGBoost).

Full-distribution zero-inflated regression. Rather than hurdle at the extreme threshold — which restricts intensity estimation to the few exceedances — we model the *entire* distribution of $\mathcal{I}$ using all wet days and derive the exceedance probability and tail from it. With a point mass $p_0(\boldsymbol{x}_t)$ at zero and a positive continuous CDF F_+,

$$F_{t+h}(z) = p_0 + (1-p_0)\,F_+(z;\boldsymbol{\theta}(\boldsymbol{x}_t))\ (z>0), \qquad \pi_{t+h} = (1-p_0)(1-F_+(u)), \tag{4}$$

with p_0 and the parameters $\boldsymbol{\theta}$ driven by linear predictors in $\boldsymbol{x}_t$. The Generalised Pareto distribution (GPD) is

$$H(y;\sigma,\xi) = 1 - (1+\xi y/\sigma)_+^{-1/\xi}\ (\xi \ne 0), \quad 1 - e^{-y/\sigma}\ (\xi = 0), \quad y \ge 0, \tag{5}$$

with $(a)_+ = \max(a,0)$ and ξ governing tail heaviness. For F_+ we compare a log-normal, a gamma, and — as the extreme-value-motivated choice — the *extended GPD* (eGPD),

$$F_+(z) = \big[H(z;\sigma,\xi)\big]^{\kappa}, \qquad \kappa > 0, \tag{6}$$

which is GPD-tailed at high values (inheriting the Pickands–Balkema–de Haan justification of extreme-value theory; Coles, 2001) yet is a single continuous model over all positive intensities, requiring *no* threshold selection (κ shapes the lower tail).

Hurdle with a GPD tail, and its deep-GP variant. The most elaborate contender is a two-part hurdle whose intensity component is a non-stationary GPD,

$$F_{t+h}(z) = (1-\pi_{t+h}) + \pi_{t+h}\,H(z-u;\sigma_{t+h},\xi), \qquad z > u, \tag{7}$$

with $\pi_{t+h} = \mathrm{sigmoid}(g_\pi(\boldsymbol{x}_t))$ and $\sigma_{t+h} = \mathrm{softplus}(g_\sigma(\boldsymbol{x}_t))$ driven by latent functions on which we place Gaussian-process priors — a deep-kernel GP (a neural feature map composed with a GP kernel; Wilson et al., 2016) or a deep GP (Damianou and Lawrence, 2013) — fitted by sparse variational inference (Salimbeni and Deisenroth, 2017). A Gaussian-likelihood GP cannot represent the zero-inflation and heavy tail of $\mathcal{I}$, which motivates the EVT likelihood. Full inference details are in Appendix B.

Dependent hurdle: a fair test of coupling. To give the elaborate model class its best chance, we also test whether *coupling* occurrence and intensity helps. The classical hurdle assumes the two are conditionally independent; physically they should be positively correlated, and coupling them would let the abundant occurrence data inform the data-starved tail. We realise this by giving (g_π, g_σ) a multi-output GP prior with an intrinsic coregionalisation (Bonilla et al., 2008; Álvarez et al., 2012), $\mathrm{Cov}(g_a(\boldsymbol{x}), g_b(\boldsymbol{x}')) = B_{ab}\, k_\theta(\boldsymbol{x}, \boldsymbol{x}')$, whose off-diagonal yields an interpretable borrowing-strength correlation $\rho = B_{\pi\sigma}/\sqrt{B_{\pi\pi}B_{\sigma\sigma}}$ ($\rho = 0$ recovers the independent hurdle). Because the excess is observed only on extreme days ρ is weakly identified, so we first study on synthetic data when the mechanism is recoverable and beneficial (Section 5.5); as we will see, it is recoverable and clearly helpful *when the dependence is strong*, but the estimator becomes biased and noisy away from that regime, which bounds how much a real-data outcome can tell us about the underlying dependence. This construction is related to correlated two-part models in biostatistics and econometrics (Olsen and Schafer, 2001; Heckman, 1979).

Implementation. Logistic regression uses L_2 regularisation ($C = 1$); the L_1 feature screen uses $C = 0.3$; XGBoost uses 300 trees, depth 4, learning rate 0.05. These are standard off-the-shelf settings, fixed *a priori* and **not tuned by cross-validation**, unlike the deep-GP, which receives a dedicated sensitivity sweep over inducing points and kernel depth (Section 5.1). We flag this asymmetry explicitly: it means the simple models are not given an implicit regularisation advantage from tuning, but a reviewer could still ask whether lightly tuning C would close part of the gap. We consider this unlikely to overturn the qualitative ordering — $C = 1$ is itself a weak, unremarkable amount of regularisation, and the deep-GP's shortfall (Section 5.1) is large and attributable to model depth rather than to a regularisation mismatch on the simpler models. The distributional regressions place linear predictors on the zero-inflation probability and the positive-part parameters and are fitted by maximum likelihood. The deep-GP–EVT hurdle uses a deep-kernel sparse variational GP ($M = 64$ inducing points, a two-hidden-layer feature map), with results averaged over five random seeds; conformal intervals target 90% coverage.

4.3 Tail-conformal calibration

Any of the models above yields a model-based predictive tail, but under misspecification its quantiles need not be calibrated. As one calibration option we assess, we wrap a model in a conformal layer (Vovk et al., 2005; Romano et al., 2019) that supplies distribution-free coverage, with two adaptations for our setting. *Tail targeting:* at the very high confidence levels relevant to extremes the required nonconformity quantile can exceed what a finite calibration set resolves (the empirical quantile saturates); when this happens we extrapolate the upper tail with a GPD fitted to the top 20% of calibration scores (a data-adaptive, not fixed-probability, exceedance threshold), following the extreme-conformal construction of Pasche et al. (2025). *Temporal adaptation:* because daily extremes are not exchangeable, we use adaptive conformal inference (Gibbs and Candès, 2021), updating the level online,

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t), \qquad \mathrm{err}_t = \mathbf{1}\{\mathcal{I}_t \notin C_t\}, \tag{8}$$

with C_t the issued interval and step size $\gamma = 0.02$, over an *expanding* calibration pool: starting from the held-out calibration set, each test-day nonconformity score is appended to the pool after it is scored, so the reference distribution for C_t grows through the test period rather than sliding over a fixed-width window. Under (8) with bounded scores the long-run coverage converges to $1 - \alpha$ irrespective of the data-generating process (Gibbs and Candès, 2021). We stress that this guarantee concerns *coverage*, not predictive quality: as Section 5.2 shows, a conformally calibrated model can still be worse than climatology on proper scores, so coverage alone is an insufficient indicator and we never report it as evidence of a good forecast.

4.4 Evaluation protocol

Because the predictability of extremes is intrinsically limited, evaluation must be both *proper* and *robust*: it must reward calibrated, sharp forecasts (maximising sharpness subject to calibration; Gneiting et al., 2007) and must resist the artefacts — single-run noise, quantile selection — that easily manufacture apparent skill in a low-signal regime. We therefore adopt the following protocol, and treat it as a contribution in its own right.

Occurrence: Brier decomposition and skill. For the exceedance probability we report the Brier score, its reliability/resolution/uncertainty decomposition (Murphy, 1973) — reliability measuring calibration, resolution measuring informativeness — and the Brier skill score against climatology. A calibrated-but-uninformative forecast (e.g. climatology) has zero resolution, so positive resolution with low reliability is the bar a useful model must clear.

Whole distribution and tail: (tw)CRPS. We score the predictive distribution with the continuous ranked probability score and its threshold-weighted form (Gneiting and Ranjan, 2011),

$$\mathrm{twCRPS}(F_{t+h}, \mathcal{I}_{t+h}) = \int_{\mathbb{R}} \left(F_{t+h}(z) - \mathbf{1}\{\mathcal{I}_{t+h} \le z\}\right)^2 w(z)\, dz, \qquad w(z) = \mathbf{1}\{z > \tau\}, \tag{9}$$

which remains proper while emphasising the tail ($\tau = u$). With $\tau = u$ the sub-threshold distribution is irrelevant, so the tail score depends only on the exceedance probability and the modelled excess.

Intensity predictability, isolated. Because the tail-weighted CRPS over all days is dominated by the many non-extreme days, we additionally isolate intensity by scoring the *excess* on observed extreme days: a CRPS of the excess (equivalently, the integral of the pinball loss over quantile levels) relative to the climatological excess distribution. The excess model is a quantile regression whose regularisation is chosen by cross-validation on the training set only, so the test set never informs model selection.

Robustness. Skill scores are computed against a climatological reference (the empirical training distribution). Every estimate carries a *moving-block* bootstrap confidence interval (1,000–2,000 resamples, block of ten extreme days, to respect short-range temporal dependence), and the intensity result is repeated across the four forward-only, expanding-window splits of Section 3.4 *and across the two cities*. The block length is not tightly identified by the short-range autocorrelation of Figure 1d alone, so, because the smallest split has only 96 test extremes in London (roughly ten blocks of ten), we checked the interval against block lengths of 5 and 20 at that split: the 90% CI moves by at most ± 0.003 around the reported $[+0.006, +0.059]$ in London and $[+0.019, +0.073]$ in Paris (Table 3), so the block-10 choice is not doing load-bearing work in the smallest-sample case. Per-feature significance is controlled for multiplicity by the Benjamini–Hochberg false-discovery rate, and feature importance is quantified by stability selection (selection frequency under subsampling). The deep-GP model is probed by a sensitivity sweep over inducing points and kernel depth. Discrimination metrics (precision–recall AUC) are reported only as secondary context, and conformal coverage is reported but, per Section 4.3, never as evidence of predictive quality.

5 Results

We evaluate every model on the 95th-percentile index of *both cities* by the proper scores of Section 4.4 against a climatological reference, with stochastic results seed-averaged and intensity skills carrying moving-block bootstrap confidence intervals. Beyond the main bake-off (Section 5.1)

Table 2: Model bake-off at the 95th percentile, $h = 1$ (seed-averaged), for both cities. Brier skill score (BSS) and tail-CRPS skill (twCRPSS) are relative to climatology; positive is better. Logistic and XGBoost are occurrence-only (no twCRPSS). The qualitative ordering is identical across cities.

	London		Paris	
Model	**BSS**	**twCRPSS**	**BSS**	**twCRPSS**
Climatology / stationary POT	0.000	0.000	0.000	0.000
Logistic (+ERA5)	**+0.038**	—	**+0.117**	—
XGBoost (+ERA5)	+0.034	—	+0.101	—
ZI gamma (+ERA5)	+0.008	−0.06	+0.090	+0.046
ZI extended-GPD (+ERA5)	+0.012	−0.045	+0.093	**+0.052**
ZI log-normal (+ERA5)	−0.29	−2.27	−0.04	−0.76
Deep-GP–EVT hurdle (TCDGP)	≈ -0.4	≈ -0.3	≈ -0.4	≈ -0.3

we report a feature screen (with false-discovery control and stability selection) identifying which covariates carry occurrence and intensity signal; the hold-out intensity test across four expanding-window splits; a deep-GP sensitivity sweep; a seed-averaged horizon sweep $h \in \{1, 3, 7, 14, 28\}$; and a synthetic study validating the dependent hurdle. Throughout, we ask whether conclusions hold across the two contrasting cities.

5.1 Simpler models win; distribution choice dominates

Table 2 reports the bake-off, and the same three findings hold in *both* cities. First, the elaborate deep-GP–EVT hurdle is *worse than climatology* on both skill scores (Brier skill ≈ -0.4 in each city), while the best occurrence models are a plain calibrated logistic regression and the gradient-boosted-tree (XGBoost) classifier — two standard, inexpensive models. Second, the choice of response distribution matters more than model class: a zero-inflated log-normal is substantially miscalibrated (tail-CRPS skill −2.3 in London), whereas the light-tailed gamma and the threshold-free extended GPD are well behaved; the index tail is lighter than a log-normal or a low-threshold Generalised Pareto fit assumes. Third, modelling the *full* zero-inflated distribution and deriving the exceedance probability from it beats hurdling at the extreme threshold, which starves the tail of data. Quantitatively everything is *stronger in Paris* (e.g. logistic Brier skill +0.117 versus London's +0.038), reflecting its larger, more convective extreme sample; indeed in Paris the extended GPD even beats climatology on the full-predictive tail score (twCRPSS +0.052), which it could not in data-scarcer London.

Is the deep-GP failure a tuning artefact? No. A sensitivity sweep over inducing points ($M \in \{32, 64, 128\}$) leaves the deep-kernel GP well below climatology in both cities (Brier skill −0.2 to −0.7). The decisive comparison is depth: replacing the deep kernel with a *plain* GP recovers a competitive model (Brier skill +0.029 in London, +0.123 in Paris) — so it is the added depth, not the Gaussian-process framework, that fails on this low-signal problem. The elaborate model is thus dominated not because GPs are unsuitable but because the extra flexibility cannot be supported by so few informative events.

Sensitivity to the exceedance threshold u. Re-fitting the winning occurrence model (logistic +ERA5) at $u \in \{0.75, 1.00, 1.25\}$ — i.e. marking a day extreme at 75%, 100%, and 125% of the qth-percentile level — leaves the qualitative finding unchanged in both cities: the Brier skill score stays clearly positive and of comparable magnitude at every u (London +0.055, +0.038, +0.028; Paris +0.103, +0.117, +0.101). In London skill rises modestly as u falls, as expected — a lower threshold admits more extreme days and so more data to fit — while Paris peaks at

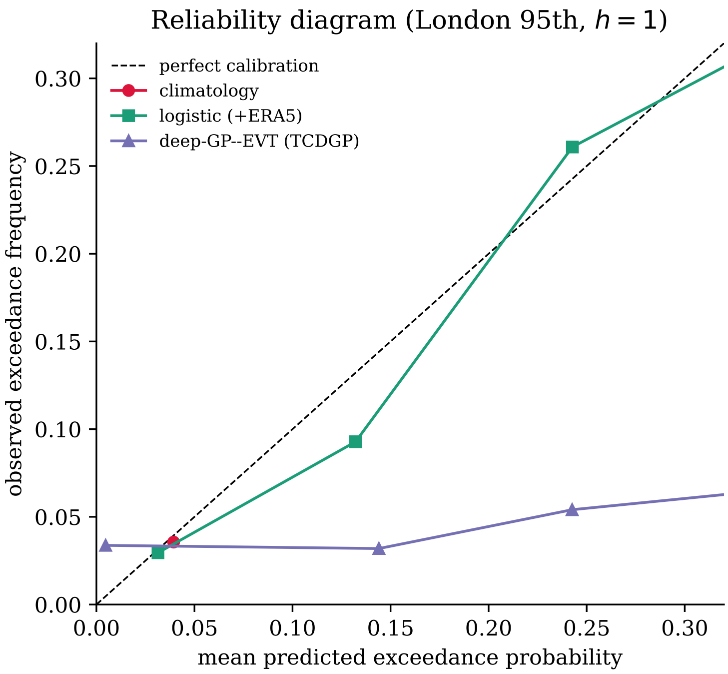

Figure 2: Reliability diagram for the exceedance probability (London 95th, h=1). A plain logistic regression tracks the diagonal (calibrated) and spreads its predictions (informative); the deep Gaussian-process extreme-value model is overconfident and nearly flat, despite its conformal intervals attaining nominal coverage. Climatology is a single point at the base rate. The Paris counterpart is Supplementary Figure S2.

$u = 1$; neither city shows the skill collapsing or changing sign near the working threshold. This is reassuring on two counts: predictability is not an artefact of the particular threshold $u = 1$, and it is not a knife-edge effect that appears at one value and vanishes at neighbouring ones.

5.2 Conformal coverage does not imply forecast quality

The tail-conformal layer attains its nominal 0.90 coverage on the real series across every predictor set and horizon (Figure 4b). But this is *guaranteed by construction* and is uninformative about predictive quality: the same deep-GP–EVT model whose conformal intervals are perfectly calibrated is simultaneously *worse than climatology* on the proper scores (Table 2). The reliability diagram (Figure 2) makes the point directly: the deep-GP–EVT model's raw exceedance probabilities are badly miscalibrated and almost flat (low resolution), whereas a plain logistic regression is well calibrated and informative. Conformal coverage should therefore not be reported as evidence of a good probabilistic forecast — a cautionary point in its own right.

5.3 Predictability of occurrence

Occurrence is weakly but genuinely predictable in both cities: the best calibrated model improves on climatology with a Brier skill of +0.038 in London and +0.117 in the more convective Paris (Table 2), with positive resolution. An L_1-regularised logistic identifies physically sensible antecedent predictors — led by the **K-index** of convective instability, the 500 hPa geopotential, integrated vapour transport, low-level humidity, and season — and stability selection confirms their robustness: the K-index and the most recent index lag are selected in essentially every training subsample (full ranking, both cities: Supplementary Table S2). Skill nonetheless decays toward the prevalence floor as the forecast horizon grows (Figure 4a; numerical BSS by horizon, both cities: Supplementary Table S3), consistent with the limited day-scale persistence of these

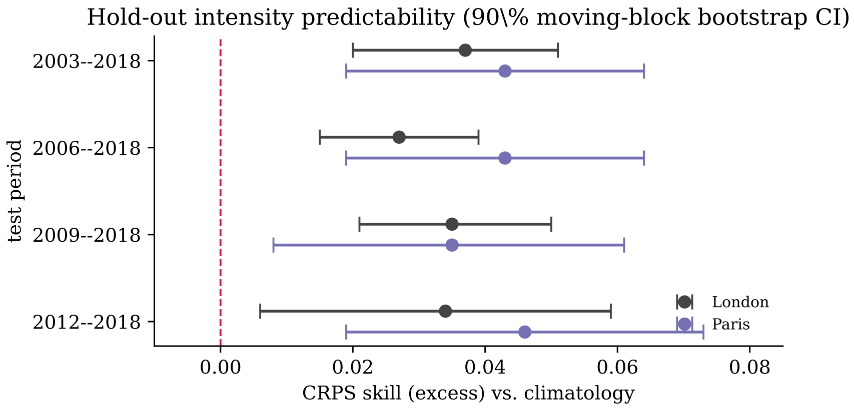

Figure 3: Hold-out intensity predictability for both cities: CRPS-of-excess skill over climatology, with 90% moving-block bootstrap intervals, across four forward-only expanding-window train/test splits. The skill is small but its interval excludes zero in every split and city; Paris point estimates run somewhat higher than London's, though the intervals overlap.

extremes. At the rarer 99th-percentile threshold (London), where the scarce data support only an occurrence analysis, the qualitative picture is unchanged (Supplementary Table S1).

5.4 Predictability of intensity

Whether the *magnitude* of an extreme is predictable is the more delicate question, and the answer is a qualified yes — in both cities. On the extreme days, many antecedent features correlate with the excess far beyond chance, and the association survives multiplicity control: 21 of 49 features in London and 27 of 49 in Paris pass a Benjamini–Hochberg false-discovery threshold (versus ~ 2 expected), led by convective inhibition and CAPE in both. To test whether this signal generalises out of sample, we fit quantile regressions of the excess on held-out extreme days (regularisation chosen by cross-validation on the training set only) and score them by the CRPS of the excess against the climatological excess distribution. Computed over a fine quantile grid on the primary split, the CRPS skill is small but its moving-block bootstrap interval *excludes zero* in each city (London $+0.014$, $[+0.009, +0.019]$; Paris $+0.036$, $[+0.005, +0.074]$). As a robustness check we recompute it over a coarser quantile grid across four independent expanding-window splits per city: the skill stays positive throughout — every interval excluding zero, with primary-split (2009–2018) values of $+0.035$ in each city (Table 3, Figure 3). The signal is concentrated in the upper tail (the largest events) and is carried by antecedent convective instability (CAPE/CIN); median magnitude is indistinguishable from climatology. The Paris point estimates run somewhat higher than London's, but their confidence intervals overlap, so we read the intensity signal as robustly *replicated* across the two cities rather than decisively stronger in either.

5.5 Synthetic validation of the dependent hurdle

On data simulated from the generative model with a *known* occurrence–intensity correlation ($\rho = 0.70$, GPD shape $\xi = 0.20$), we verify that the coregionalised hurdle recovers the dependence and that it helps. The *coupled* model recovers $\widehat{\rho} \approx +0.83$, while the *independent* hurdle (the $\rho\!=\!0$ special case) cannot. Crucially, the independent model's intensity head — estimated from the few exceedances alone — becomes unstable (its shape collapses so that test excesses fall outside the GPD support), whereas coupling lets the abundant occurrence data stabilise the tail and yields orders-of-magnitude better tail log-likelihood. This is direct evidence for the borrowing-strength mechanism: the dependent hurdle matters most exactly where intensity data are scarcest.

Table 3: Hold-out intensity predictability in both cities: CRPS-of-excess skill over climatology across forward-only expanding-window splits, with 90% moving-block bootstrap intervals. Every interval excludes zero; Paris point estimates run somewhat higher than London's, though the intervals overlap.

Test period	**# test ext.**	**CRPS skill**	90% **CI**	**P**(>0)
London				
2003–2018	213	+0.037	[+0.020, +0.051]	1.00
2006–2018	175	+0.027	[+0.015, +0.039]	1.00
2009–2018	130	+0.035	[+0.021, +0.050]	1.00
2012–2018	96	+0.034	[+0.006, +0.059]	0.97
Paris				
2003–2018	329	+0.043	[+0.019, +0.064]	1.00
2006–2018	275	+0.043	[+0.019, +0.064]	1.00
2009–2018	211	+0.035	[+0.008, +0.061]	0.99
2012–2018	164	+0.046	[+0.019, +0.073]	0.99

Misspecification. Because the validation above fixes a single ρ, we repeat it (three seeds each) at $\rho_{\text{true}} \in \{0.3, 0.7, 0.9\}$ with $\xi = 0.20$ held fixed. Recovery is reliable only near the validated setting: at $\rho_{\text{true}} = 0.7$ the coupled model recovers $\widehat{\rho} = 0.81 \pm 0.07$ (matching the single-seed estimate above), but at $\rho_{\text{true}} = 0.3$ and 0.9 the estimate is both biased and far noisier ($\widehat{\rho} = 0.61 \pm 0.46$ and 0.60 ± 0.17 respectively) — which is consistent with, and now direct evidence for, the paper's own diagnosis that ρ is weakly identified from so few exceedances (Section 4.2). The borrowing-strength benefit is correspondingly uneven across seeds: large tail-log-likelihood gains appear in most but not all fits, and shrink to near-zero at $\rho_{\text{true}} = 0.9$, where the independent model no longer reliably collapses. The qualitative conclusion — coupling never does worse, and in most seeds does much better, than the independent hurdle — holds throughout the sweep, but the *size* of the benefit, like $\widehat{\rho}$ itself, should be read as a favourable single-setting illustration rather than a precisely quantified effect.

What does the real data say? Fitting the coupled model on the real series (five seeds) gives $\widehat{\rho} = +0.25 \pm 0.14$ in London and $\widehat{\rho} = +0.64 \pm 0.12$ in Paris. These cannot be read directly as the strength of real occurrence–intensity dependence, for the reason the misspecification sweep just exposed: in this data regime the estimator is biased toward mid-range positive values, returning $\widehat{\rho} \approx 0.6$ even when the true correlation is only 0.3. A real-data $\widehat{\rho}$ near 0.6 is therefore consistent with anything from weak to moderate dependence, and we do not use it to claim either presence or absence of exploitable coupling. Crucially, the substantive conclusion does not depend on estimating ρ: the coupled deep-GP–EVT hurdle fails to beat climatology in both cities (Table 2), and the depth sensitivity of Section 5.1 localises that failure to the deep kernel's excess flexibility — a plain GP is competitive — rather than to the coupling mechanism. The shortfall is thus one of model capacity given so few informative events; whether a lower-capacity coupled model could convert any real dependence into skill is a question this experiment leaves open. We deliberately stop short of the cleaner-sounding claim that the real index simply lacks exploitable dependence, because the misspecification result shows we cannot measure that dependence reliably enough to assert its absence.

5.6 Skill decays with horizon

Both occurrence skill and discrimination decay toward the climatological floor as the forecast horizon grows (Figure 4a), consistent with the limited day-scale persistence of London extremes; conformal coverage, being guaranteed by construction, is flat across horizons (Figure 4b). Seed-

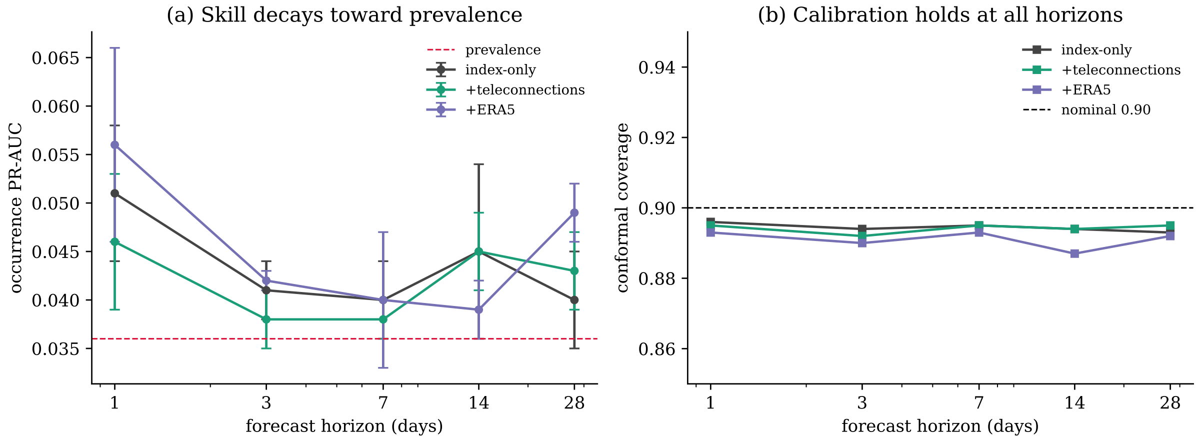

Figure 4: Seed-averaged horizon sweep on the London 95th-percentile index. **(a)** Occurrence PR-AUC decays toward the prevalence floor (dashed) as the forecast horizon grows; driver gains are marginal at all horizons (the apparent ERA5 edge at $h = 28$ does not survive a higher-seed check; see text). **(b)** Conformal coverage stays at the nominal 0.90 across all horizons and predictor sets. Error bars / markers are means over three seeds. The Paris counterpart is Supplementary Figure S3.

averaged sweeps found no statistically robust occurrence-discrimination gain from atmospheric drivers at any single horizon — a reminder that the predictability documented above is weak and easily masked by noise unless assessed with seed-averaging and proper scores rather than single runs. The same decay and flat coverage hold for Paris (Supplementary Figure S3).

Summary. Across the bake-off, simpler calibrated models dominate the elaborate deep-GP–EVT model, and the response distribution matters more than the model class (Section 5.1); conformal coverage holds but is, on its own, uninformative about forecast quality (Section 5.2). The substantive finding is a small but *rigorously established* predictability: occurrence is weakly predictable from antecedent convective-instability and circulation indices (Section 5.3), and the magnitude of the largest events is weakly predictable from antecedent convective instability, with a CRPS skill over climatology that excludes zero across four independent splits (Section 5.4). The dependent-hurdle mechanism is separately validated on synthetic data (Section 5.5). Every one of these conclusions holds in *both* London and Paris, and is generally more pronounced in the more convective Paris — evidence that they reflect the meteorology of precipitation extremes rather than the idiosyncrasies of one city.

6 Discussion

Predictability is real but small, and physically interpretable. The headline is not a forecasting triumph but an honest measurement: in both cities, the occurrence of a daily precipitation extreme and the magnitude of the largest events carry weak antecedent predictability, on the order of a few percent of CRPS skill over climatology. Such effect sizes are modest but not unusual for precipitation at these lead times — sub-seasonal precipitation forecast skill is itself typically small — and their practical value lies less in a sharp point forecast than in *trustworthy, calibrated* probabilities, plus a slim but real edge for anticipating the most intense events. That the signal is led by convective-instability indices (the K-index for occurrence; CAPE and convective inhibition for intensity) is reassuring: it is the thermodynamics one would expect to govern intense convective rainfall, it argues against the skill being an artefact, and it is precisely why the effects are *larger in the more convective Paris*. The median magnitude of an extreme, by contrast, is indistinguishable from climatology: once an exceedance occurs, its size is largely unpredictable except in the upper tail. One nuance reinforces the data-dependence of these statements: in data-scarcer London no model beats the empirical climatological tail on the full-predictive score, but in Paris — with half again as many extremes — the extended GPD does, so "the tail is climatology-bound" is a sample-size effect, not a universal law.

Methodological lessons. Three are worth emphasising. First, *model complexity did not pay*: in both cities a deep Gaussian-process extreme-value hurdle was beaten by logistic regression, by gradient-boosted trees, and by simple zero-inflated distributional regressions. A sensitivity analysis localises the failure to the added *depth* — a plain Gaussian process is competitive, but the deep kernel cannot be supported by so few informative events; Gaussian processes earn their keep through structure (notably spatial structure), which a single-site temporal problem does not provide. Second, the *response distribution dominates*: a log-normal tail was substantially miscalibrated, and hurdling at the extreme threshold starved the tail of data; modelling the full zero-inflated distribution with a threshold-free extended GPD was far better. Third, and most relevant to the wider literature, *conformal coverage is insufficient as a stand-alone indicator of forecast quality*: it is attained by construction even by a model worse than climatology on proper scores. This is a known property of coverage, but our experiments make it concrete — reporting coverage without a proper-score comparison can create a false impression of a good probabilistic forecast.

One predictive distribution, not two analyses. We report occurrence and intensity separately for interpretability, but they are not disconnected: they are the point-mass and tail components of a single hurdle predictive distribution, $F(z) = (1 - \pi) + \pi\, H(z - u)$, with occurrence probability $\pi(\mathbf{x})$ and a covariate-dependent excess law H. Because the (unstructured) hurdle likelihood factorises in these parameters, fitting the two components jointly is *equivalent* to fitting them separately — they are orthogonal pieces of one distribution rather than two ad hoc models — and a single threshold-weighted CRPS over F decomposes additively into an occurrence and an intensity contribution. Fitting this unified hurdle directly and scoring it end-to-end (Appendix C) recovers exactly the two-part story: in both cities the overall skill splits into a modest occurrence term and a small but unusually stable intensity term (its bootstrap interval excludes zero in both cities), with negligible interaction. The only way to make the joint fit differ from the separate one is to *couple* the components through shared latent structure — precisely the dependent deep-GP hurdle we tested. It did not improve on the separable treatment here; but, as Section 5.5 makes clear, that outcome reflects the limited capacity supportable from so few exceedances (and an occurrence–intensity correlation we cannot reliably estimate), not a demonstration that no coupling could ever help. The separable treatment is thus the pragmatic, well-identified choice at this data volume, rather than a refutation of dependence in principle.

Why rigour mattered here. Several apparent results did not survive scrutiny: a single-seed driver benefit and an apparent long-horizon gain both evaporated under seed-averaging, and an across-the-board intensity skill shrank to an upper-tail effect once regularisation was chosen without peeking at the test set. The predictability we report is what remained after seed-averaging, proper scoring, cross-validated model selection, bootstrap intervals, and multi-split robustness. In a low-signal regime these safeguards are not optional — they are the difference between a finding and an artefact.

Limitations. The study covers two cities — a step beyond a single site, but still a limited sample of European urban climates, so we frame the results as a careful two-city assessment rather than a continent-wide claim. London and Paris also share a climate zone, a downscaled reanalysis product (ERA5-2km), and a data provider (Copernicus), so they are not fully independent replicates: common reanalysis biases could in principle inflate both cities' apparent skill together. Replication in a climatically distinct setting (Mediterranean or continental-interior) would be needed to rule this out, and we flag it as the most direct next step rather than a claim we can settle here. It uses a perfect-prognosis design (antecedent reanalysis fields, not operational forecasts), and the intensity analysis conditions on observed exceedances, so it isolates predictability rather than delivering an end-to-end warning. The extreme-day definition itself has two, distinct knobs — the standardisation percentile q (Section 3.1) and the exceedance threshold u applied to the standardised index (Section 4.1) — and we vary each separately (the 99th-percentile analysis in the Supplementary Material, and the u-sensitivity check of Section 5.1) rather than conflating them. We also treat the 1989–2018 climatological reference as stationary; given that climate projections indicate a rising trend in extreme rainfall over this period (IPCC, 2021), a non-stationary reference (e.g. a smoothly time-varying baseline rate) could in principle sharpen or shift the skill scores reported here, and we did not pursue it because the sample is already short relative to the rarity of the events, making an additional non-stationary component hard to fit reliably; we flag it as a natural extension rather than a design choice we defend as final. Finally, the index is itself an ERA5-derived precipitation product, which raises a circularity concern. We address it in two ways: no precipitation field of any kind enters the predictors, and every driver is strictly lagged, so the predictors are antecedent dynamical and thermodynamic state (pressure, geopotential, humidity, instability), never the rainfall itself. Using antecedent state to predict future precipitation is the standard perfect-prognosis setting; the residual caveat — that a reanalysis is internally consistent across variables and time — cannot make a strictly-lagged, non-precipitation predictor circular, but is worth keeping in mind, and would be removed entirely by moving to numerical-weather-prediction forecast drivers (future work).

7 Conclusion

We set out to measure, rather than merely to model, the predictability of rare precipitation extremes, and to do so with an evaluation protocol robust to the pitfalls of a low-signal setting. The answer, consistent across two climatically contrasting cities, is a qualified, quantified "a little": occurrence and the magnitude of the largest events are weakly but genuinely predictable from antecedent convective instability, while simpler calibrated models suffice and conformal coverage is no substitute for a proper-score comparison. That every finding holds in — and is generally more pronounced in — the more convective Paris is itself evidence that these are properties of precipitation extremes, not of one city. Natural extensions are the directions in which more signal, and a genuine role for structured Gaussian-process models, are most plausible: more cities to map the predictability across climates, *spatial* modelling across the grid of cells, *operational* drivers from numerical weather prediction in place of reanalysis, and *sub-seasonal* aggregations where slowly varying drivers may contribute. Whether predictability rises materially in those settings is, in the spirit of this paper, a question to be settled by proper scores and confidence intervals rather than asserted.

Data and code availability

The underlying data are publicly available: the London and Paris precipitation-extreme indices derive from the Copernicus Climate Data Store product *Extreme precipitation risk indicators for Europe and European cities (1950–2019)*, and the atmospheric predictors from ERA5 (Copernicus CDS, `reanalysis-era5-single-levels`; and the WeatherBench2 regridded ERA5 archive) and from the NOAA teleconnection indices. The code accompanying this paper — implementations of all models (logistic, gradient-boosted trees, the zero-inflated log-normal/gamma/eGPD distributional regressions, and the deep-GP–EVT hurdle), the full evaluation protocol (proper scores, the hold-out intensity test, moving-block bootstrap and multi-split robustness), the data-retrieval and feature-processing scripts, and the scripts that produce every figure and table — together with the processed daily feature tables for both cities, are publicly available at `https://github.com/olatunjijohnson/precip-extremes-predictability`. Following journal policy, the processed data files will additionally be deposited on figshare with a citable DOI upon acceptance.

Author contributions

Fankai Dong: methodology, software, formal analysis, investigation, visualisation, writing — original draft. **Olatunji Johnson:** conceptualisation, methodology, supervision, writing — review and editing. *[Please confirm the CRediT roles reflect the actual division of work.]*

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Conflict of interest

The authors declare no conflict of interest. *[Please confirm.]*

A Data and predictors

Response. The standardised daily precipitation-extreme index for London (Section 3), derived from the ERA5-2km downscaled reanalysis; we use the daily maximum over the 53 London grid cells, with an extreme defined by $\mathcal{I} > 1$, at the 95th (primary) and 99th percentile standardisations.

Antecedent predictors. All predictors are functions of information up to the forecast origin t (a horizon-h forecast of day $t + h$ uses day-t information only); no precipitation field is used, since the response is itself an ERA5-derived precipitation product.

- **Index history:** lags $\mathcal{I}_{t-1}, \ldots, \mathcal{I}_{t-14}$; rolling mean over the previous 7 and 30 days; rolling maximum over 7 days; count of exceedances in the previous 30 days; and sin / cos harmonics of day-of-year.
- **ERA5 circulation and moisture** (WeatherBench2, 1.5°, daily means over a city-centred synoptic box, $\sim 15° \times 30°$ — e.g. 45–60°N, 20°W–10°E for London, shifted south-east for Paris): mean sea-level pressure and its meridional/zonal gradients, total-column water vapour, 2 m temperature, 10 m wind components and speed, geopotential at 500 and 850 hPa, and specific humidity at 850 hPa.
- **ERA5 convective instability** (Copernicus CDS, `reanalysis-era5-single-levels`, 0.25°, daily mean and maximum over the box): CAPE, convective inhibition (CIN), the K-index, the Total-Totals index, total-column water vapour, the vertical-integral eastward/northward water-vapour-flux components and their magnitude (IVT).
- **Teleconnection indices** (monthly, broadcast to daily): North Atlantic Oscillation and Arctic Oscillation (NOAA CPC), and Niño 3.4 (NOAA PSL).

Retrieval. WeatherBench2 fields were read directly from the public Google Cloud zarr store with `xarray`; the convective-instability fields were retrieved from the Copernicus CDS via `cdsapi` with server-side spatial subsetting to the box, 6-hourly sampling aggregated to daily statistics; teleconnection indices were downloaded from NOAA. Every driver is forward-filled to daily resolution where needed and lagged by the forecast horizon. Processing code and the cached daily feature table accompany the paper.

B The deep-GP extreme-value hurdle: inference

We give the inference for the most elaborate contender (Section 4.2). The parameters depend on covariates through latent functions, $\pi = \mathrm{sigmoid}(g_\pi(\boldsymbol{x}_t))$, $\sigma = \mathrm{softplus}(g_\sigma(\boldsymbol{x}_t))$, and a constant shape ξ_0 (covariate-dependent ξ is an optional extension). For horizon h the per-day likelihood factorises into occurrence and (on extreme days) intensity terms,

$$p(O_t, \mathcal{I}_t \mid \boldsymbol{g}(\boldsymbol{x}_{t-h})) = \pi^{O_t}(1-\pi)^{1-O_t} \left[h_{\mathrm{GPD}}(\mathcal{I}_t - u; \sigma, \xi_0)\right]^{O_t} \left[f^{\mathrm{blw}}(\mathcal{I}_t)\right]^{1-O_t}, \tag{10}$$

with h_{GPD} the GPD density and f^{blw} a zero-inflated positive density for the sub-threshold regime.

Sparse variational inference. We place GP priors on $\boldsymbol{g} = (g_\pi, g_\sigma)$ and introduce $M \ll N$ inducing variables $\boldsymbol{u} = \boldsymbol{g}(\boldsymbol{Z})$ with whitened variational posterior $q(\boldsymbol{u}) = \mathcal{N}(\boldsymbol{m}, \boldsymbol{S})$. The evidence lower bound is

$$\mathcal{L} = \sum_t \mathbb{E}_{q(\boldsymbol{g}(\boldsymbol{x}_{t-h}))}\big[\log p(O_t, \mathcal{I}_t \mid \boldsymbol{g}(\boldsymbol{x}_{t-h}))\big] - \mathrm{KL}\big(q(\boldsymbol{u}) \,\|\, p(\boldsymbol{u})\big), \tag{11}$$

the expectation computed by Monte Carlo over the marginal $q(\boldsymbol{g}(\boldsymbol{x}_{t-h}))$. Inducing-point inference reduces cost to $\mathcal{O}(NM^2)$; the kernel acts on a neural feature map (deep kernel) or, for the deep GP, on a composition of GP layers fitted by doubly-stochastic variational inference (Salimbeni and Deisenroth, 2017). Hyperparameters, inducing inputs/distribution, neural weights, and ξ_0 are learned jointly by maximising (11).

Coregionalised (dependent) hurdle. We couple the two latents through a linear model of coregionalisation: with Q shared base GPs $f_q \sim \mathcal{GP}(0, k_\theta)$, $g_a(\boldsymbol{x}) = \sum_q W_{aq} f_q(\boldsymbol{x})$, so $\mathrm{Cov}(g_a, g_b) = B_{ab}\, k_\theta$ with $\boldsymbol{B} = \boldsymbol{W}\boldsymbol{W}^\top$. The correlation $\rho = B_{\pi\sigma}/\sqrt{B_{\pi\pi}B_{\sigma\sigma}}$ is the borrowing-strength parameter; a diagonal $\boldsymbol{W}$ recovers the independent hurdle. Because the excess is observed only on extreme days, ρ is weakly identified, so we use a weakly-informative prior shrinking $\rho \to 0$ and validate recovery on synthetic data (Section 5.5).

Tail-conformal layer. For an interval at level $1-\alpha$ we use a conformalised-quantile-regression nonconformity score on the model's predictive quantiles, $E_t = \max\{\widehat{z}_{\alpha/2} - \mathcal{I}_t,\ \mathcal{I}_t - \widehat{z}_{1-\alpha/2}\}$; when the required score quantile exceeds what the calibration set resolves we extrapolate with a GPD fitted to the top 20% of the upper calibration scores (Pasche et al., 2025), and we adapt the level online by (8) (step size $\gamma = 0.02$) over an expanding calibration pool that accumulates test-day scores as the walk-forward evaluation proceeds.

C Unified hurdle: one predictive distribution, decomposed

To verify that the separately reported occurrence and intensity analyses are the two components of one inferential object (Section 6), we fit a single hurdle predictive distribution directly,

$$F(z \mid \boldsymbol{x}) = \begin{cases} 1 - \pi(\boldsymbol{x}) & z \le u, \\ (1 - \pi(\boldsymbol{x})) + \pi(\boldsymbol{x})\, H(z - u; \sigma(\boldsymbol{x}), \xi) & z > u, \end{cases}$$

with $\pi(\boldsymbol{x}) = \mathrm{sigmoid}(\boldsymbol{a}^\top \boldsymbol{x})$, $\sigma(\boldsymbol{x}) = \mathrm{softplus}(\boldsymbol{b}^\top \boldsymbol{x})$, H the GPD excess law and ξ a covariate-free tail index, fitting all parameters by a *single* joint hurdle log-likelihood. Because that likelihood factorises, this is numerically equivalent to the separate fits; the point is the *evaluation*. We score the whole F with one threshold-weighted CRPS at $\tau = u$ and attribute its skill over a parametric climatology (constant base rate $\bar{\pi}$ plus a stationary GPD tail) by swapping one component at a time: S_{occ} uses $\pi(\boldsymbol{x})$ with the climatological tail, S_{int} uses $\bar{\pi}$ with $\sigma(\boldsymbol{x})$, and S_{full} uses both; the decomposition is additive up to a small interaction. Intervals are moving-block bootstrap (90%, block = 10 days, seed-averaged over three fits).

twCRPS skill vs climatology	London	Paris
S_{full} (occurrence + intensity)	+0.030 [+0.002, +0.061]	+0.084 [+0.027, +0.135]
occurrence contribution S_{occ}	+0.029 [+0.009, +0.050]	+0.075 [+0.043, +0.104]
intensity contribution S_{int}	+0.007 [+0.005, +0.009]	+0.013 [+0.010, +0.017]
occurrence $\times$ intensity interaction	−0.005	−0.004
occurrence BSS (unified π / standalone logistic)	+0.036 / +0.038	+0.116 / +0.117

Table 4: The single hurdle distribution, scored end-to-end, decomposes into the two findings reported in the main text: a modest occurrence term and a small but stable intensity term (its bootstrap interval excludes zero in both cities), with negligible interaction. The unified occurrence probability reproduces the standalone logistic skill, confirming the components are orthogonal pieces of one distribution. The intensity contribution is small because it is averaged over all test days; restricted to the $\sim$3–6% of days that are extreme it matches the isolated CRPS-of-excess skill of Section 5.4.

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